

China Solid-State Battery Industry Quarterly Data Briefing — Q2 2026

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What's Inside

本期关键词

2026 designated as the **mass production year** for semi-solid-state batteries

H1 investment exceeds RMB 50 billion; full-sector planned investment surpasses RMB 50 billion

2025 semi-solid-state battery installations reach 31.7 GWh, up 272% YoY

2026 market size projected to reach **RMB 31.8 billion**

Solid-state battery consumption tax exemption from September 2026 through end-2028

Global solid-state battery shipments expected to reach **607.2 GWh by 2030**

Three numbers that matter

2025 semi-solid-state battery installations reached 31.7 GWh, up 272% year-over-year, with penetration climbing to 4.1%. 2026 penetration is projected to rise further to 5.7%. The industry has crossed the critical threshold from "sample to product."

2026 H1 solid-state battery sector planned investment exceeds RMB 50 billion, with over 40 public investment projects. Solid-state electrolyte investment has grown approximately **4× year-over-year**, signaling a decisive shift toward upstream material self-sufficiency.

From September 2026 through end-2028, solid-state batteries will be exempt from consumption tax. For a 20 GWh production line, this exemption saves tens of millions of yuan annually in tax costs — a critical buffer during the early production ramp-up phase.

Part 1: Market Overview

1.1 Industry Positioning

2026 is widely recognized as the **mass production year** for semi-solid-state (mixed solid-liquid) batteries. The industry has transitioned from “concept validation” to “pre-mass-production” — a critical accumulation phase marked by intensive conferences, standards issuance, technological breakthroughs, capital inflows, and capacity rollouts. The second half of the year will enter the “sprint” stage, with multiple vehicle models equipped with semi-solid-state batteries scheduled for 集中上市.

The current market presents a clear structure: **semi-solid-state batteries entering large-scale commercial application, while all-solid-state batteries**

undergo small-batch validation. All-solid-state batteries are expected to enter pilot installation in 2027, with volume production still requiring time.

1.2 Market Size

China's solid-state battery market reached **RMB 19.7 billion in 2025**, up 107.4% year-over-year. Industry analysts project the 2026 market size will reach **RMB 31.8 billion**.

Global perspective: Global solid-state battery shipments reached approximately **17.9 GWh in Q1 2026**, with a market size of approximately RMB 1.163 billion. Full-year 2026 global shipments are projected to reach **50 GWh**, with a market size of approximately **RMB 47 billion**, representing year-over-year growth exceeding 213%.

Long-term outlook: By 2030, global solid-state battery shipments could reach **607.2 GWh**, of which all-solid-state batteries are expected to account for nearly 20%. The China solid-state automotive battery market is forecast to grow from a pilot-scale phase in 2026 to a commercial production value exceeding **USD 8–12 billion by 2035**, representing a compound annual growth rate of approximately **55–70%**.

1.3 Shipment & Installation Data

Shipment trends:

2023–2025: Semi-solid-state battery shipments grew from 2.5 GWh to 10 GWh

2026E: Shipments projected to exceed **14 GWh**

Installation trends:

2025: Semi-solid-state battery installations reached **31.7 GWh**, up **272% YoY**

2025 penetration: **4.1%**

2026E penetration: Projected to reach **5.7%**

Capacity outlook: GGII projects 2026 China solid-state battery shipments will exceed the **15 GWh level** — a more than 10× increase from the GWh-level volumes of 2025. This represents the industry’s transition from “laboratory concept” to “production line reality”.

1.4 Application Scenarios

Solid-state batteries’ future market space will primarily come from expanded application scenarios, including:

Application	2026 Outlook
New Energy Vehicles	Primary application; flagship models and high-end passenger vehicles 优先 adopt
eVTOL (Electric Vertical Take-Off and Landing)	Market space projected to reach RMB 9.5 billion in 2026
Humanoid Robots	10Ah all-solid-state batteries already developed for this segment

Application	2026 Outlook
Energy Storage Systems	Emerging application; demonstration projects underway
Aerospace & Satellite	Satellite demand estimated at tens to hundreds of GWh annually

Semi-solid-state batteries will first be used in **high-performance scenarios** such as low-altitude economy and premium passenger vehicles, then gradually 渗透 into large-capacity LFP battery application areas. Liquid lithium-ion batteries (including semi-solid-state) will remain the absolute mainstream for power batteries through 2030.

Part 2: Investment & Financing

2.1 Overall Investment Scale

H1 2026: The solid-state battery sector saw over **40 public investment projects**, with total planned investment across the entire sector exceeding **RMB 50 billion**.

Solid-state electrolyte was the fastest-growing segment in H1 2026, with planned investment exceeding **RMB 8 billion** — approximately a **4×** **increase** from the sub-RMB 2 billion investment scale of H1 2025. This reflects the industry's strategic shift toward upstream material self-sufficiency.

Semi-solid-state batteries: H1 2026 related planned investment reached **RMB 4 billion**, maintaining steady year-over-year growth.

2.2 Notable Investment Projects

Company / Project	Investment	Capacity	Status
Tuoyi Solid Energy	RMB 6 billion	30 GWh	Located in Inner Mongolia; three-phase rollout
Qingtao Energy	—	30 GWh	Kunshan + Inner Mongolia combined capacity
Hohhot Solid-State Project	RMB 800 million	2.2 GWh	Production line + R&D pilot line; Jun 2026–Jun 2027
Guoxiang Century	—	10 GWh+	Shanghai solid-state complex

2.3 Financing Activity

Notable 2026 financings:

Company	Round	Amount	Details
Qingtao Energy	Series H	~RMB 1.9 billion	Completed February 2026; filed for Hong Kong IPO; cumulative 11 financing rounds
Zhibang Lithium	Series A	Undisclosed	May 2026; lead investor Nongfushanquan; ~10% stake
Liyuanheng	Private	Up to RMB	SSE review; targets solid-state

Company	Round	Amount	Details
	Placement	1.618B	battery equipment R&D
Shen'an Lithium Energy	Pre-A	~RMB 100M	July 2026; investors: Huaxia Capital, Shengshi Investment, Changzhe Investment

2.4 H2 2026 Investment Momentum

H2 2026 new industry investment has already exceeded **RMB 12 billion**, with multiple production lines compatible with semi-solid-state technology entering the commissioning phase.

Part 3: Policy & Regulatory Developments (Q2 2026)

3.1 Consumption Tax Exemption (Critical Policy)

On July 17, the Ministry of Finance, General Administration of Customs, and State Taxation Administration jointly issued the Announcement on Adjusting the Consumption Tax Policy for Certain Batteries.

Key provisions:

From **September 1, 2026 through December 31, 2028**, the following battery types are exempt from consumption tax:

Sodium-ion batteries

Solid-state batteries